EXPERIMENT : 14

AIM:

Experiment to understand feature scaling.

DESCRIPTION:

Understand the importance of feature scaling.

CODE:

```
import numpy as np
import pandas as pd

# Read dataset
df = pd.read_csv('/content/Data.csv')

# Display dataset
print(df)
print(df.head())

# Fill missing Country values with mode
df['Country'] = df['Country'].fillna(df['Country'].mode()[0])

# Separate features and label
features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values

# Handle missing Age and Salary using mean
from sklearn.impute import SimpleImputer

age = SimpleImputer(strategy='mean')
salary = SimpleImputer(strategy='mean')

features[:, [1]] = age.fit_transform(features[:, [1]])
features[:, [2]] = salary.fit_transform(features[:, [2]])

print(features)
```

```
# One-Hot Encoding for Country
from sklearn.preprocessing import OneHotEncoder

oh = OneHotEncoder(sparse_output=False, handle_unknown='ignore')
country = oh.fit_transform(features[:, [0]])

print(country)

# Combine encoded Country, Age and Salary
final_set = np.concatenate(
    (country, features[:, [1, 2]].astype(float)), axis=1
)

print(final_set)

# Standardization
from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
feat_standard_scaler = sc.fit_transform(final_set)

print("Standardized data:")
print(feat_standard_scaler)

# Normalization using MinMaxScaler
from sklearn.preprocessing import MinMaxScaler

mms = MinMaxScaler(feature_range=(0, 1))
feat_minmax_scaler = mms.fit_transform(final_set)

print("Normalized data:")
print(feat_minmax_scaler)
```

OUTPUT:

```
  Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0      NaN       Yes
5   France  35.0  58000.0       Yes
```

```
6    Spain   NaN  52000.0        No
7   France  48.0  79000.0       Yes
8  Germany  50.0  83000.0        No
9   France  37.0  67000.0       Yes
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0      NaN       Yes
[['France' 44.0 72000.0]
 ['Spain' 27.0 48000.0]
 ['Germany' 30.0 54000.0]
 ['Spain' 38.0 61000.0]
 ['Germany' 40.0 63777.77777777778]
 ['France' 35.0 58000.0]
 ['Spain' 38.77777777777778 52000.0]
 ['France' 48.0 79000.0]
 ['Germany' 50.0 83000.0]
 ['France' 37.0 67000.0]]
[[1. 0. 0.]
 [0. 0. 1.]
 [0. 1. 0.]
 [0. 0. 1.]
 [0. 1. 0.]
 [1. 0. 0.]
 [0. 0. 1.]
 [1. 0. 0.]
 [0. 1. 0.]
 [1. 0. 0.]]
[[1.00000000e+00 0.00000000e+00 0.00000000e+00 4.40000000e+01
  7.20000000e+04]
 [0.00000000e+00 0.00000000e+00 1.00000000e+00 2.70000000e+01
  4.80000000e+04]
 [0.00000000e+00 1.00000000e+00 0.00000000e+00 3.00000000e+01
  5.40000000e+04]
 [0.00000000e+00 0.00000000e+00 1.00000000e+00 3.80000000e+01
  6.10000000e+04]
 [0.00000000e+00 1.00000000e+00 0.00000000e+00 4.00000000e+01
  6.37777778e+04]
 [1.00000000e+00 0.00000000e+00 0.00000000e+00 3.50000000e+01
  5.80000000e+04]
 [0.00000000e+00 0.00000000e+00 1.00000000e+00 3.87777778e+01
  5.20000000e+04]
```

```
 [1.00000000e+00 0.00000000e+00 0.00000000e+00 4.80000000e+01
  7.90000000e+04]
 [0.00000000e+00 1.00000000e+00 0.00000000e+00 5.00000000e+01
  8.30000000e+04]
 [1.00000000e+00 0.00000000e+00 0.00000000e+00 3.70000000e+01
  6.70000000e+04]]
Standardized data:
[[ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01  7.58874362e-01
   7.49473254e-01]
 [-8.16496581e-01 -6.54653671e-01  1.52752523e+00 -1.71150388e+00
  -1.43817841e+00]
 [-8.16496581e-01  1.52752523e+00 -6.54653671e-01 -1.27555478e+00
  -8.91265492e-01]
 [-8.16496581e-01 -6.54653671e-01  1.52752523e+00 -1.13023841e-01
  -2.53200424e-01]
 [-8.16496581e-01  1.52752523e+00 -6.54653671e-01  1.77608893e-01
   6.63219199e-16]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01 -5.48972942e-01
  -5.26656882e-01]
 [-8.16496581e-01 -6.54653671e-01  1.52752523e+00  0.00000000e+00
  -1.07356980e+00]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01  1.34013983e+00
   1.38753832e+00]
 [-8.16496581e-01  1.52752523e+00 -6.54653671e-01  1.63077256e+00
   1.75214693e+00]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01 -2.58340208e-01
   2.93712492e-01]]
Normalized data:
[[1.         0.         0.         0.73913043 0.68571429]
 [0.         0.         1.         0.         0.        ]
 [0.         1.         0.         0.13043478 0.17142857]
 [0.         0.         1.         0.47826087 0.37142857]
 [0.         1.         0.         0.56521739 0.45079365]
 [1.         0.         0.         0.34782609 0.28571429]
 [0.         0.         1.         0.51207729 0.11428571]
 [1.         0.         0.         0.91304348 0.88571429]
 [0.         1.         0.         1.         1.        ]
 [1.         0.         0.         0.43478261 0.54285714]]
```

RESULT:

Feature scaling was successfully performed on the given dataset using appropriate scaling techniques. It helped bring numerical features to a similar scale, improving the performance of machine learning algorithms.